

Dr Hussain Ahmad

Cyber Security | Artificial Intelligence | Software Engineering

Adelaide, Australia | +61 420565253 | [Email](#) | [Website](#) | [Institutional Profile](#) | [Google Scholar](#) | [LinkedIn](#)

Profile Highlights

- Expertise in AI-driven Cybersecurity, AI Security, Generative AI, Agentic AI, Software Engineering, Information Technology, Cloud Computing, Data Science, and Self-Adaptive Systems.
- Led 40+ R&D projects in collaboration with industry and academic partners, resulting in 25+ research papers, including publications in CORE A*/A venues, with 1000+ citations and an h-index of 13.
- Supervised more than 80 undergraduate and postgraduate students on industry-focused R&D projects.
- Secured R&D funding from prominent enterprises, including the Department of State Development South Australia and Cyber Security Cooperative Research Centre.
- Responsible for the curriculum design, coordination, and delivery of undergraduate and postgraduate courses across Cybersecurity, Artificial Intelligence, and Data Science, achieving a 99% student experience rating.
- Awarded prestigious awards for exceptional achievements in R&D, high-quality teaching, industry collaboration, leadership, professional and community service.
- Contributed to the global research community as a reviewer, program committee member, and editorial board member for leading international journals and conferences.
- Served as an ambassador and mentor for postgraduate researchers, advocating for their interests and concerns, and representing Equity, Diversity, and Inclusion within the higher-degree research community.

Education

Doctor of Philosophy (PhD)

May 2021 – July 2025

[The University of Adelaide, Australia](#)

Thesis Title: Towards AI-Driven, Self-Adaptive, and Cyber-Resilient Software Engineering [[Abstract View](#)]

Thesis outcome: Distinguished Research Thesis - Accepted without Corrections [[link](#)]

Advisors: [Prof. Markus Wagner](#); [Prof. Claudia Szabo](#); [Prof. Christoph Treude](#)

Bachelor of Science in Electronic Engineering

Aug 2013 – May 2017

[Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Pakistan](#)

Accredited Qualification by [Engineers Australia](#); Grade: High Distinction

Work Experience

Lecturer – Cybersecurity, Artificial Intelligence, Software Engineering

Feb 2025 – Present

[Adelaide University, Australia](#) (formerly [University of Adelaide](#))

- **Research:** Conduct research in cybersecurity and artificial intelligence, with a focus on AI-enabled cyber defence, cyber risk and situational awareness, agentic AI, and large language models.
- **Research Publications:** Prepare manuscripts using LaTeX, manage submissions and revisions, address reviewer feedback, and publish in peer-reviewed journals and international conferences.
- **R&D Collaboration:** Contribute to R&D collaborations across government, defence, industry, and startups, including [Defence Trailblazer](#), [CSIRO's Data61](#), and [Stone & Chalk](#), advancing cyber and AI solutions.
- **Research Funding:** Identify and pursue competitive research funding opportunities, develop grant proposals, build collaborative funding partnerships, and contribute to funding applications.
- **Student Supervision:** Supervise undergraduate and postgraduate students on industry-focused R&D projects, providing technical and inclusive feedback, directing their research efforts, and evaluating their projects.
- **Teaching:** Design and deliver curricula for Cybersecurity and AI courses, integrating industry practices and interactive pedagogy to enhance student engagement, critical thinking, and problem-solving.
- **Professional & Community Service:** Serve as an editorial board member, program committee member, and reviewer for international journals and conferences.

Doctoral Researcher – *AI-driven Software Security & Scalability*

May 2021 – July 2025

The University of Adelaide, Australia

- Led 15+ R&D projects at the intersection of Cybersecurity, AI, and Software Engineering, resulting in 12 research papers with publications in top-tier conferences and journals.
- Secured industry funding by collaborating with global and Australian industry partners, aligning research with stakeholder priorities, and delivering high-impact R&D deliverables.
- Advocated for researchers as a Postgraduate Researchers Representative and Gender, Equity, Diversity, and Inclusion (GEDI) Representative, fostering an inclusive and supportive academic environment.

Project Manager (*Voluntary*)

Nov 2023 – Dec 2024

*Migrova - An immigration marketplace*Partners: *Stone & Chalk, Antler*

- Provided expert consultation in Cybersecurity and Artificial Intelligence for ICT projects.
- Contributed to securing A\$100,000 Seed-Start Grant from the *Department of State Development (SA)*.
- Delivered the funded project, deploying AI-enabled system to improve operational efficiency.

Software Engineer (*Voluntary*)

Dec 2022 – Feb 2023

Kindship - NDIS service provider

Developed a *Machine Learning-Enabled Therapist Recommendation Engine* to match the requirements of differently abled children with the right therapists on the Kindship platform.

Research Assistant

July 2020 – Apr 2021

The University of Adelaide, Australia

Funded Project (A\$422,000): Enhancing Cyber Defense Mechanisms in Command-and-Control Systems

Partners: *Cyber Security Cooperative Research Centre; Defense Science and Technology Group; Cisco*

- Identified and reported vulnerabilities, cyber-attacks, and countermeasures in command-and-control systems.
- Simulated cyber-attacks to evaluate tactical impacts and assess system resilience.
- Conducted static and dynamic risk assessments, devising self-adaptive mechanisms to mitigate security risks.

Process Worker

Jan 2020 – May 2020

Prolife Foods, Australia - Great Food with Obsessive Service

Responsible for quality and production assurance of roaster machine operations as per defined Standard Operating Procedures (SOPs) and Key Performance Indicators (KPIs).

Process Worker

Dec 2019 – Jan 2020

Kelly Services, Australia - Connect people to work in ways that enrich their lives

Worked through Kelly Services on short-term assignments with various organizations, including *VistaPrint*, where I prepared and packed customer orders, and *Kuehne+Nagel*, where I supported shipment preparation and dispatch.

Tutor

Aug 2018 – Nov 2019

Private Tuition Centre, Pakistan

Taught Mathematics and Physics to students across Years 5-12, adapting lessons to different learning levels. Supported students in strengthening their subject knowledge, problem-solving skills, and academic performance.

Trainee Engineer

July 2017 – July 2018

Tetra Pak, Pakistan - World's leading food processing and packaging solutions company

Led a team of 20+ operators across three shifts for the Laminator Machine, managing and reporting KPIs. Actively contributed to automating manual processes for enhanced time and cost efficiency, aligning with Industry 4.0 goals.

Research Grants

- Contributed to securing an **AUD 100,000 Seed-Start Grant** from the [Department of State Development South Australia](#) to enhance the digital operations of [Migrova](#) through AI-driven technology integration. (2024)
- Secured a **Seed Grant** from the [School of Computer Science](#), University of Adelaide, to advance focused research on **Security for AI** with end-to-end AI pipeline security. (2025)
- Awarded **AUD 75,000 Research Grant** from the [Cyber Security Cooperative Research Centre](#) to advance research on [Cyber Common Operating Picture](#) platforms. (2021)
- Awarded **competitive funding** through the [CSIRO ON Prime Innovation Program](#) to support customer discovery, market validation, and research commercialization. (2025)
- Secured **AUD 7,000** by [TriumphED](#) to develop and deploy an AI-driven personalized course and mentor recommendation system aimed at improving student engagement, retention, and academic outcomes. (2025)
- Awarded a competitive **travel grant** to travel to Macau, China, and present my research at the [32nd Asia-Pacific Software Engineering Conference](#). (2025)
- Awarded **University of Adelaide Scholarship** valued at **AUD 28,597** per annum, providing financial support to undertake full-time doctoral research in cybersecurity and artificial intelligence.
- Awarded **AUD 5,000 Google Cloud Grant** to support R&D on microservice auto-scaling operations. (2024)
- Recipient of **AUD 15,000 Financial Assistance Award** by **GIKI** for four-year undergraduate studies. (2013)

Awards & Achievements

- Honoured to be nominated for the **STEM Award** in the 7NEWS Young Achiever Awards Australia. [\[link\]](#)
- Attained an exceptional **99% Student Experience of Learning and Teaching (SELT) rating** for teaching and coordinating courses at the University of Adelaide. [\[link\]](#)
- Featured by the University of Adelaide for my PhD journey in recognition of outstanding academic and research achievements. [\[link\]](#)
- Awarded **Outstanding International Student Award** at the University of Adelaide for excellence in R&D, impactful industry collaborations, and exceptional leadership during my PhD. [\[link\]](#)
- Featured in a leading newspaper on the global stage for receiving the Outstanding International Student Award at the University of Adelaide. [\[link\]](#)
- PhD thesis at the University of Adelaide **accepted without corrections** following international expert review, reflecting exceptional originality, rigor, and global research excellence. [\[link\]](#)
- Featured in a leading international newspaper highlighting my PhD achievement in Cyber Security at the University of Adelaide. [\[link\]](#)
- Awarded **Exceptional HDR Representative Award** for outstanding advocacy and community engagement in support of postgraduate researchers. [\[link\]](#)
- Awarded both the **People's Choice Award** and **Second Place** in the 2024 Visualise Your Thesis Competition at the University of Adelaide, recognizing the impact and innovation of my R&D activities. [\[link\]](#)
- Featured by the University of Adelaide for my academic journey from Bachelor's student → PhD graduate → Lecturer, with the aim of motivating and guiding current and prospective university students. [\[link\]](#)
- Accredited as a **Professional Electronics Engineer** by [Engineers Australia](#).
- Awarded six **Dean's Excellence Awards** in undergraduate studies.

Teaching Portfolio

Teaching Philosophy – 99% Student Experience Rating

- 1- Structured Learning:** Develop a modular pedagogical approach that supports progressive concept development.
- 2- Simplicity with Rigour:** Make complex ideas accessible while maintaining technical depth.
- 3- Real-World Relevance:** Integrate industry-aligned, hands-on learning to connect theory with practice.

4- Constructive Feedback: Provide continuous, targeted feedback to support student growth and confidence.

5- Inclusivity & Engagement: Foster inclusive, student-centered learning through interactive, flexible approaches.

Lecturer (Unit Instructor/Coordinator) – *On-Campus, Hybrid, and Online Delivery*

- Cloud Security (INFO 6004) [[link](#)]
- Concepts in Cyber Security (COMP SCI 7328) [[link](#)]
- Introduction to System Security (COMP 5059) [[link](#)]
- Research Methods for Cyber Security (COMP 5063) [[link](#)]
- Big Data Analysis and Project (COMP SCI 7209) [[link](#)]
- Working with Big Data (INFO 6017) [[link](#)]
- Foundations of Computer Science - Python B (COMP 5058) [[link](#)]
- OL Research Methods (COMP 5062) [[link](#)]
- OL Human and Ethical Factors in Computer Science (COMP 6029) [[link](#)]
- Innovative and Emerging Technologies (COMP 5032) [[link](#)]
- Enterprise Resource Management (INFO 6008) [[link](#)]
- Industry Project in Information Technology (COMP SCI 3021 & COMP SCI 3021MELB) [[link](#)]
- Professional Communication and Teamwork (INFO 2032) [[link](#)]

Student Supervision

- Supervised 80+ undergraduate and postgraduate students across research and industry capstone projects in cybersecurity, AI, data science, and software engineering.
- Mentored students in research design, critical thinking, and academic writing, with several projects leading to research publications [[1](#), [2](#), [3](#), [4](#), [5](#)].
- Led industry-linked projects in collaboration with external partners, enabling real-world problem-solving, system design, and stakeholder coordination [[link](#)].

Some of the key research projects supervised are outlined below:

- 1- Cyber Security Industry Project (COMP SCI 7102A & COMP SCI 7102B)
 - Advancing Cybersecurity Vulnerability Assessment with Large Language Models
 - Microservice Vulnerability Analysis: A Literature Review with Empirical Insights
 - Towards Deep Learning Enabled Cybersecurity Risk Assessment for Microservice Architectures
 - Explainable Autonomous Cyber Defense using Adversarial Multi-Agent Reinforcement Learning
 - SecureChat: Enhancing Real-Time Messaging Security using RSA Cryptography
- 2- Artificial Intelligence and Machine Learning Research Project (COMP SCI 7205A & COMP SCI 7205B)
 - Agentic AI for Software Security
 - Advancing Cybersecurity through AI-Powered Deception Techniques
 - Secure IoT Trustworthy AI-driven Resilient Adaptive Mechanism
 - AI-Driven Scalability and Security in Microservice Architectures
 - Design and Development of Australian Vulnerability Database
- 3- Research Project Master Computer Science (COMP SCI 7100A & COMP SCI 7100B)
 - Multi-agent Intervention in Neurodevelopmental Disorder System
 - AI Pipeline Security Framework for Detection Assessment and Mitigation
 - Optimizing Prompt Engineering through Evolutionary Computation
- 4- Master of Computing & Innovation Project (COMP SCI 7098)
 - A Comparative Study of AI Models for Recommendation Systems
 - Horizontal Pod Auto-scaling of Software Systems Using Multi-Metric Optimization
 - A Self-Adaptive Framework for Software Security

- 5- Industry Project in Information Technology (COMP SCI 3021)
 - GenAI Policy Scanner: Mapping and Supporting Ethical GenAI Use in Australasian Universities
 - AI-Powered Recommendation Platform for Study Pathways and Mentorship
 - AI-Driven Migration Services in Australia

Curriculum Development

- Cloud Security (INFO 6004) [\[link\]](#)
- Concepts in Cyber Security (COMP SCI 7328) [\[link\]](#)
- Big Data Analysis and Project (COMP SCI 7209) [\[link\]](#)
- Industry Project in Information Technology (COMP SCI 3021 & COMP SCI 3021MELB) [\[link\]](#)

Research Portfolio

Publications – 1000+ Citations, h-index: 13 (collaborative research with government, defence, and industry partners)

1. **Hussain Ahmad**, Isuru Dharmadasa, Faheem Ullah, and M. Ali Babar. "A Review on C3I Systems' Security: Vulnerabilities, Attacks, and Countermeasures." *ACM Computing Surveys* (2023). **Impact Factor 30.4** [\[link\]](#)
Partners: [Defense Science and Technology Group](#); [Cyber Security Cooperative Research Centre](#); [Cisco](#)
2. **Hussain Ahmad**, Christoph Treude, Markus Wagner, Claudia Szabo. "Smart HPA: A Resource-Efficient Horizontal Pod Auto-scaler for Microservice Architectures." *21st IEEE International Conference on Software Architecture* (2024). **CORE Rank A** [\[link\]](#)
Partners: [The University of Melbourne](#); [Monash University](#)
3. **Hussain Ahmad**, Christoph Treude, Markus Wagner, Claudia Szabo. "Towards Resource-Efficient Reactive Proactive Auto-scaling for Microservices." *Journal of Systems and Software* (2025). **Impact Factor 4.1** [\[link\]](#)
Partners: [Monash University](#); [Singapore Management University](#)
4. **Hussain Ahmad**, Christoph Treude, Markus Wagner, Claudia Szabo. "Resilient Auto-Scaling of Microservice Architectures with Efficient Resource Management." *32nd IEEE Asia-Pacific Software Engineering Conference* (2025). [\[link\]](#)
Partners: [Monash University](#); [Singapore Management University](#)
5. **Hussain Ahmad**, Faheem Ullah, and Rehan Jafri. "A Survey on Immersive Cyber Situational Awareness Systems." *Journal of Cybersecurity and Privacy* (2025). **Impact Factor 3.8** [\[link\]](#)
Partners: [Honeywell, UK](#); [Zayed University, UAE](#)
6. Yiyao Zhang, Diksha Goel, **Hussain Ahmad**. "Explainable Autonomous Cyber Defense using Adversarial Multi-Agent Reinforcement Learning." *Expert Systems with Applications* (2026). **Impact Factor 9.4** [\[link\]](#)
Partner: [CSIRO's Data61](#)
7. Majid Abdulsatar, **Hussain Ahmad**, Diksha Goel, Faheem Ullah. "Towards Deep Learning Enabled Cybersecurity Risk Assessment for Micro Architectures." *Cluster Computing* (2025). **Impact Factor 5.5** [\[link\]](#)
Partners: [CSIRO's Data61](#); [Zayed University, UAE](#)
8. Raveen Jayalath, **Hussain Ahmad**, Diksha Goel, Shuja Syed, and Faheem Ullah. "Microservice Vulnerability Analysis: A Literature Review with Empirical Insights." *IEEE Access* (2024). **Impact Factor 4.2** [\[link\]](#)
Partners: [CSIRO's Data61](#); [Migrova](#); [SLB - Schlumberger, USA](#)
9. Farwa Abbas, **Hussain Ahmad**, Claudia Szabo. "SCALAR: Self-Calibrating Adaptive Latent Representation Learning." *37th IEEE International Conference on Tools with Artificial Intelligence* (2025). [\[link\]](#)
Partner: [Imperial College London](#)
10. Asrul Arifin, **Hussain Ahmad**, Yiyao Zhang, Diksha Goel. "AgenticVM: Agentic AI for Adaptive Software Vulnerability Management." *IEEE Software* (2026). **Impact Factor 4.0** [\[link\]](#)
Partner: [CSIRO's Data61](#)
11. Mubin Ul Haque, Isuru Dharmadasa, Zarrin Sworna, Roshan Rajapakse, **Hussain Ahmad**. "I think this is the most disruptive technology: Exploring Sentiments of ChatGPT Early Adopters." (2023). [\[link\]](#)

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12. Shivansh Chopra, **Hussain Ahmad**, Diksha Goel, Claudia Szabo. "ChatNVD: Advancing Cybersecurity Vulnerability Assessment with Large Language Models." *IEEE Access* (2026). **Impact Factor 4.2** [\[link\]](#)
Partner: [CSIRO's Data61](#)
 13. Yiyao Zhang, Diksha Goel, **Hussain Ahmad**, Claudia Szabo. "RegimeFolio: A Regime Aware ML System for Sectoral Portfolio Optimization in Dynamic Markets." *IEEE Access* (2025). **Impact Factor 4.2** [\[link\]](#)
Partner: [CSIRO's Data61](#)
 14. Subin Santhosh, Farwa Abbas, **Hussain Ahmad**, Claudia Szabo. "Comparative Analysis of Large Language Models in Healthcare." submitted to *Journal of Healthcare Informatics Research* (2026). [\[link\]](#)
Partner: [SAHMRI - South Australian Health and Medical Research Institute](#)
 15. Diksha Goel, **Hussain Ahmad**, Kristen Moore, Mingyu Guo. "Co-Evolutionary Defence of Active Directory Attack Graphs via GNN-Approximated Programming." minor revision at *Expert Systems with Applications* (2025). [\[link\]](#)
Partner: [CSIRO's Data61](#)
 16. Yiyao Zhang, Diksha Goel, **Hussain Ahmad**, Jun Shen. "VACS: Value-Aligned Compositional Shielding for Multi-Agent Reasoning." submitted to *Decision Support Systems* (2026). [\[link\]](#)
Partners: [University of Wollongong](#); [CSIRO's Data61](#)
 17. **Hussain Ahmad** and Diksha Goel. "The Future of AI: Exploring the Potential of Large Concept Models." submitted to *Journal of Information and Telecommunication* (2026). [\[link\]](#)
Partner: [CSIRO's Data61](#)
 18. Farwa Abbas, **Hussain Ahmad**. "Robust Partial Least Squares Using Low and Sparse Decomposition." (2024). [\[link\]](#)
Partner: [Imperial College London](#)
 19. Diksha Goel, **Hussain Ahmad**, Ankit Jain, Nikhil Goel. "Machine Learning Driven Smishing Detection Framework for Mobile Security." submitted to *Journal of Software: Evolution and Process* (2025). [\[link\]](#)
Partner: [CSIRO's Data61](#)
 20. Yiyao Zhang, Diksha Goel, **Hussain Ahmad**, Shixun Huang, Jun Shen. "CausalNav: Reliability-Certified Causal World Models for Control under Parameter Shift." submitted to *Information Fusion* (2026). [\[link\]](#)
Partners: [University of Wollongong](#); [CSIRO's Data61](#)
 21. Faheem Ullah, Xiaohan Ye, Uswa Fatima, Zahid Akhtar, Yuxi Wu, **Hussain Ahmad**. "What Skills Do Cyber Security Professionals Need?" *Information and Computer Security* (2026). [\[link\]](#)
Partners: [Zayed University UAE](#); [State University of New York Polytechnic Institute, USA](#)
 22. Tanvi Jois, **Hussain Ahmad**, Fatima Noor, Faheem Ullah. "Australian Bushfire Intelligence with AI-Driven Environmental Analytics." submitted to *Natural Hazards* (2026). [\[link\]](#)
Partner: [Zayed University, UAE](#)
 23. Yiyao Zhang, Diksha Goel, **Hussain Ahmad**, Jun Shen. "Beyond Reactive Agents: Uncertainty-Gated Meta-Reasoning for Tool-Augmented Decision-Making." submitted to *Neurocomputing* (2026). [\[link\]](#)
Partners: [University of Wollongong](#); [CSIRO's Data61](#)
 24. Yiyao Zhang, Diksha Goel, **Hussain Ahmad**, Shixun Huang, Jun Shen, Aniket Mahanti, Ankit Jain. "Symbolically Checked Multi-Step Reasoning via Verification-Weighted Search." submitted to *Journal of Intelligent Information Systems* (2026). [\[link\]](#)
Partners: [University of Auckland](#); [University of Wollongong](#); [CSIRO's Data61](#)
 25. Kefan Chen, **Hussain Ahmad**, Diksha Goel, Claudia Szabo. "3S-Trader: A Multi-LLM Framework for Stock Selection via Scoring and Iterative Strategy." submitted to *Expert Systems with Applications* (2025). [\[link\]](#)
Partner: [CSIRO's Data61](#)
 26. Yiyao Zhang, Diksha Goel, **Hussain Ahmad**, Jun Shen. "Beyond Reactive Agents: Uncertainty-Gated Meta-Reasoning for Tool-Augmented Decision-Making." submitted to *Frontiers in Computer Science* (2026). [\[link\]](#)
Partners: [University of Wollongong](#); [CSIRO's Data61](#)

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27. Yiyao Zhang, Diksha Goel, **Hussain Ahmad**, Shixun Huang, Jun Shen. "When the Judge Should Not Decide: Evidence-Locked, Non-Compensatory Selection Bounds LLM-Judge Failure in Reasoning Pipelines." submitted to *Machine Learning with Applications* (2026). [[link](#)]
Partners: [University of Wollongong](#); [CSIRO's Data61](#)
 28. **Hussain Ahmad**, Claudia Szabo, Troy Boswell, Toan Le. "Self-Adaptive Architecture for Autonomous Decision-Making in Contested Environments." *in progress* (2026).
Partners: [Defence Trailblazer](#); [DEWC Services](#)

Industry R&D Projects

1. **Hussain Ahmad**, Faheem Ullah, and Muhammad Ali Babar. "Simulation Study for Analyzing the Security of C3I Systems." submitted to *Cyber Security Cooperative Research Centre* (2021).
Partners: [Defense Science and Technology Group](#); [Cyber Security Cooperative Research Centre](#); [Cisco](#)
2. **Hussain Ahmad**, Faheem Ullah, and Muhammad Ali Babar. "Static and Dynamic Security Risk Analysis for C3I Systems." submitted to *Cyber Security Cooperative Research Centre* (2021).
Partners: [Defense Science and Technology Group](#); [Cyber Security Cooperative Research Centre](#); [Cisco](#)
3. **Hussain Ahmad**, Faheem Ullah, and Muhammad Ali Babar. "Self-Adaptive Security Risk Management Framework for C3I Systems." submitted to *Cyber Security Cooperative Research Centre* (2021).
Partners: [Defense Science and Technology Group](#); [Cyber Security Cooperative Research Centre](#); [Cisco](#)
4. Zhenyang Li, Xiao Han, Jiazhi Chen, Yiu Tam, Kun Lei, Ho Li, **Hussain Ahmad**. "Smart Migration: Machine Learning-Based Agent Recommendation System." submitted to *Department of State Development South Australia* (2024).
Partners: [Department of State Development](#); [CSIRO](#); [Deakin University](#); [Migrova](#)
5. Emelia Oliver, Brooke Seigert, Het Thakkar, **Hussain Ahmad**. "Causal Connectivity Mapping of Working Memory Networks." submitted to [SAHMRI - South Australian Health and Medical Research Institute](#) (2025).
6. Lehao Huang, Hanyi Xiao, Hanyu Li, **Hussain Ahmad**. "Design and Prototype an Embedded Rewards & Benefits Engine for Gig Platforms." submitted to [MyGigsters](#) (2025).
7. James Nguyen, Sukhmandeep Kaur, Akshar Brahmabhatt, Shaik Abdul Hadi, Pranshubhai Mehta, **Hussain Ahmad**. "Mobility Assessment Portal." submitted to [Add-Life](#) (2025).
8. Kim Hyuna, Shuxin Zhang, Qishuo Zheng, Junlong Li, **Hussain Ahmad**. "AI-Powered Recommendation Platform for Study Pathways and Mentorship." submitted to [TriumphED](#) (2025).
9. Lucas Caruso, Hok Lam Lo, Ge Wang, Tomoki Nonogaki, **Hussain Ahmad**. "AI for Safer Cyberspace: Detecting Harmful or Misleading Online Content." submitted to [CSIRO's Data61](#) (2025).
10. Michael King, Christian Nesci, Nicholas Bologiannis, John-Paul Chamings, **Hussain Ahmad**. "Advancing AUSWO Migration Platform." submitted to [AUSWO](#) (2025).
11. Danial El-Kaboult, Matthew Pirone, George Vagionas, **Hussain Ahmad**. "Image Processing Software for Serial Number Identification." submitted to [SLB \(Schlumberger\)](#) (2025).
12. Mitchell David, Kelly Wibowo, Gia Thinh, **Hussain Ahmad**. "Closed-Loop Brain Stimulation Controller Using Machine Learning." submitted to [SAHMRI - South Australian Health and Medical Research Institute](#) (2025).
13. Jack Jim, Shuaichen Yang, Sankalp Waghe, **Hussain Ahmad**. "A Self-Adaptive Framework for Web Application Development and Management." submitted to [Aphrodite's Harvest](#) (2026).
14. Andrew Jones, Alen Joyce, Athul Krishna, **Hussain Ahmad**. "Development of a Digital Membership Management System for the Brazilian Association of South Australia." submitted to [Brazilian Association of South Australia](#) (2026).
15. Nok Chan, Rashmeet Kaur, Sahaana Suganya, **Hussain Ahmad**. "Data-Driven Retail Optimization for Multi-Brand Performance Analysis." submitted to [Bickford's Group](#) (2026).

Presentations – *International Conferences, Industry Forums, Academic Seminars*

1. Presented the research "SCALAR: Self-Calibrating Adaptive Latent Attention Representation Learning." at *37th IEEE International Conference on Tools with Artificial Intelligence, Greece* (2025). [[link](#)]
2. Presented the research "Resilient Auto-Scaling of Microservice Architectures with Efficient Resource Management." at *32nd Asia-Pacific Software Engineering Conference, China* (2025). [[link](#)]
3. Presented PhD research on self-adaptive software security and scalability at the *Visualise Your Thesis Competition, University of Adelaide* (received the People's Choice Award and Second Place) (2024). [[link](#)]
4. Invited Talk: "Smart HPA: A Resource-Efficient Horizontal Pod Auto-scaler for Microservice Architectures." at *CSIRO's Data61, Melbourne* (2024). [[link](#)]
5. Invited Talk: "Self-Adaptive Software Security and Scalability." at *Maptek, Adelaide* (2024). [[link](#)]
6. Invited Talk: "Secure Coding Principles." at *UoA Cyber Security Awareness Seminar, Adelaide* (2023). [[link](#)]
7. Presented the research "A Review on C3I Systems' Security: Vulnerabilities, Attacks, and Countermeasures." at *Cyber Security Cooperative Research Centre Symposium, Melbourne* (2022). [[link](#)]
8. Presented the research "Secure Multi-User Content Sharing for Augmented Reality Applications." at *Centre for Research on Engineering Software Technologies, Adelaide* (2022). [[link](#)]

Media Articles

1. The risks of AI dependence for developing nations. *The News International, Pakistan* (2025). [[link](#)]
2. Digital growth vs cyber threats: The critical need for cybersecurity. *Daily Times, Pakistan* (2024). [[link](#)]
3. Importance of cybersecurity education in Pakistan. *The News International, Pakistan* (2024). [[link](#)]

Professional & Community Service

Editorial Board Member – *Manage peer review; ensure quality and integrity*

- Journal of Intelligence & Robotics (Jan 2026 – Present) [[link](#)]
- Journal of Cybersecurity and Privacy (Aug 2025 – Present) [[link](#)]

Program Committee Member – *Review papers; support selection decisions; uphold quality standards*

- ICSA Main Research Track (July 2025 – Feb 2026) [[link](#)]
- ICSA Artifact Evaluation Track (Feb 2024 – Mar 2024) [[link](#)]

Reviewer – *Research Conferences & Journals*

July 2025 – Present

Reviewed manuscripts for *Journal of Cybersecurity and Privacy*, *IEEE Internet of Things Journal*, *Cluster Computing*, *Concurrency and Computation: Practice and Experience*, and *International Conference on Software Architecture*.

HDR Representative

Dec 2023 – Dec 2024

Served as the Representative of Postgraduate Researchers, advocating their concerns and interests while also organizing engaging meetups, to cultivate a positive, constructive, and supportive research environment. [[link](#)]

GEDI Representative

Jan 2024 – Dec 2024

Held the position of Gender, Equity, Diversity, and Inclusion (GEDI) Representative for Higher Degree Researchers, dedicated to fostering a diverse, equitable, and inclusive environment for everyone. [[link](#)]

Student Mentor

Jan 2024 – Sept 2024

Mentored new PhD students, helping them address challenges they encounter during their research journey. [[link](#)]

Professional Credentials

ICT Security Specialist - *Australian Computer Society (ACS)*

Received a positive Australian skills assessment, reflecting expertise in cybersecurity, vulnerability assessment, security analysis, cyber risk management, and information systems security. [\[link\]](#)

Software Engineer - *Australian Computer Society (ACS)*

Received a positive Australian skills assessment, reflecting capabilities in software architecture, system design, software development, testing, integration, and implementation. [\[link\]](#)

Software & Applications Programmer - *Australian Computer Society (ACS)*

Received a positive Australian skills assessment, reflecting capabilities in programming, application development, debugging, testing, software maintenance, and technical documentation. [\[link\]](#)

University Lecturer - *VETASSESS*

Received a positive Australian skills assessment, reflecting experience in university teaching, course development and coordination, research, student supervision, assessment design, and academic engagement. [\[link\]](#)

Electronics Engineer - *Engineers Australia*

Received a positive Australian skills assessment, reflecting capabilities in electronic systems, circuit design and analysis, embedded systems, hardware integration, testing, and engineering problem-solving. [\[link\]](#)

Professional Memberships & Registration

- Australian Information Security Association (AISA) - Member | ID: 32850 [\[link\]](#)
- Australian Computer Society (ACS) - Member | ID: 4484225 [\[link\]](#)
- Association for Computing Machinery (ACM) - Member | ID: 0338926 [\[link\]](#)
- Association for the Advancement of Artificial Intelligence (AAAI) - Member | ID: 656994 [\[link\]](#)
- Engineers Australia (EA) - Registered | ID: 9878231 [\[link\]](#)
- Institute of Electrical and Electronics Engineers (IEEE) - Member | ID: 100465884 [\[link\]](#)
- IEEE Computer Society - Member | ID: 100465884 [\[link\]](#)
- IEEE Young Professionals - Member | ID: 100465884 [\[link\]](#)
- Pakistan Engineering Council (PEC) - Registered Engineer until Dec 2021 | ID: ELECTRO/27572 [\[link\]](#)

Skills

- **Teaching Skills:** Curriculum Design, Course Delivery, Assessment & Feedback, Student Engagement, Active Learning, Project Supervision, Academic Mentoring, Inclusive Teaching, Industry Integration, Pedagogy, LMS (MyLearning), Mentimeter, Online Labs (RangeForce).
- **Technical Skills:** Python (Pandas, NumPy, scikit-learn), AI/ML & Deep Learning (TensorFlow, PyTorch), Reinforcement Learning, LLMs & Agentic AI, Prompt Engineering & RAG, Explainable AI (XAI), Secure Software Engineering & DevSecOps, Vulnerability Scanning & Penetration Testing (Trivy, Snyk, Nmap, Nessus, OWASP ZAP), Cloud-Native & Kubernetes Security (Docker, Kubernetes, AWS EKS/EC2), Incident Response & Security Operations, IoT & Cyber-Physical Systems Security, Frameworks (MITRE ATT&CK, NIST, FAIR, MAPE-K), Data Visualisation (Matplotlib, Plotly), Network Simulators (ns2, NetSim).
- **Root Cause Analysis:** 5W1H, Fault Tree Analysis, Failure Mode and Effects Analysis, Pareto Analysis, Kaizen.
- **Interpersonal Skills:** Project Management, Teamwork, Supervision, Consultation, Liaison, Event Management.

References

Professor Claudia Szabo – *Acting Dean*

School of Computer Science and Information Technology, Adelaide University, Australia

Email: claudia.szabo@adelaide.edu.au

Professor Markus Wagner – *PhD Supervisor*

Department of Data Science and AI, Monash University, Australia

Email: markus.wagner@monash.edu

Dr Diksha Goel – *Industry Project Partner*

Research Scientist, CSIRO's Data61, Australia

Email: diksha.goel@csiro.au